

HDOC DAY - 26

ALGORITHM:

Main Program:

1. Start.
2. Declare choice.
3. Display the menu:
 - Right-aligned increasing triangle
 - Right-aligned increasing number triangle
 - Right-aligned decreasing triangle
 - Exit
4. Input the user's choice.
5. Use a switch statement:
 - If choice is 1, call `increasingTriangle()`.
 - If choice is 2, call `numberTriangle()`.
 - If choice is 3, call `decreasingTriangle()`.
 - If choice is 4, exit the program.
 - Otherwise, display "Invalid choice".
6. Repeat the menu until the user selects Exit.
7. Stop.

Function 1: Right-aligned increasing triangle

1. Input the number of rows `n`.
2. Check whether `n` is positive.
3. Use an outer loop from `i = 1` to `n`.
4. Print `n-i` groups of spaces before the stars.
5. Print `i` asterisks in the row.
6. Move to the next line.

Function 2: Right-aligned increasing number triangle

1. Input the number of rows n.
2. Check whether n is positive.
3. Use an outer loop from i = 1 to n.
4. Print the required spaces to right-align the row.
5. Use an inner loop from j = 1 to i.
6. Print the value of j.
7. Move to the next line.

Function 3: Right-aligned decreasing triangle

1. Input the number of rows n.
2. Check whether n is positive.
3. Use an outer loop from i = n down to 1.
4. Print the required leading spaces.
5. Print i asterisks.
6. Move to the next line.

C PROGRAM:

```
#include <stdio.h>
```

```
/* Function declarations */
```

```
void increasingTriangle();
```

```
void numberTriangle();
```

```
void decreasingTriangle();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
printf("\nRIGHT-ALIGNED PATTERNS\n");  
printf("1. Right-aligned increasing triangle\n");  
printf("2. Right-aligned increasing number triangle\n");  
printf("3. Right-aligned decreasing triangle\n");  
printf("4. Exit\n");
```

```
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
switch (choice)  
{  
    case 1:  
        increasingTriangle();  
        break;  
  
    case 2:  
        numberTriangle();  
        break;  
  
    case 3:  
        decreasingTriangle();  
        break;  
  
    case 4:  
        printf("Exiting program...\n");  
        break;  
  
    default:
```

```

        printf("Invalid choice.\n");
    }

} while (choice != 4);

return 0;
}

/* Right-aligned increasing triangle */
void increasingTriangle()
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }

    for (i = 1; i <= n; i++)
    {
        /* Print spaces */
        for (j = 1; j <= n - i; j++)
        {

```

```
        printf(" ");
    }

    /* Print stars */
    for (j = 1; j <= i; j++)
    {
        printf("* ");
    }

    printf("\n");
}

/* Right-aligned increasing number triangle */
void numberTriangle()
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid number of rows.\n");
        return;
    }
}
```

```
for (i = 1; i <= n; i++)
{
    /* Print spaces */
    for (j = 1; j <= n - i; j++)
    {
        printf(" ");
    }

    /* Print numbers */
    for (j = 1; j <= i; j++)
    {
        printf("%d ", j);
    }

    printf("\n");
}
}
```

```
/* Right-aligned decreasing triangle */
void decreasingTriangle()
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
```

```

{
    printf("Invalid number of rows.\n");
    return;
}

for (i = n; i >= 1; i--)
{
    /* Print spaces */
    for (j = 1; j <= n - i; j++)
    {
        printf(" ");
    }

    /* Print stars */
    for (j = 1; j <= i; j++)
    {
        printf("* ");
    }

    printf("\n");
}
}

```

TEST CASE TABLE:

TEST CASE	MENU CHOICE	INPUT(n)	EXPECTED RESULT
1	1	4	Increasing right-aligned triangle with 4 rows
2	2	4	Increasing number triangle from 1 to 4

3	3	4	Decreasing right-aligned triangle with 4 rows
4	1	1	Prints one *
5	2	1	Prints 1
6	3	1	Prints one *
7	1	0	Displays Invalid number of rows.
8	2	-3	Displays Invalid number of rows.
9	5	--	Displays Invalid choice.
10	4	--	Exits the program

```
(kali㉿kali)-[~]  
$ ./h26
```

RIGHT-ALIGNED PATTERNS

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit

Enter your choice: 1

Enter the number of rows: 4

```
      *  
     * *  
    * * *  
   * * * *
```

RIGHT-ALIGNED PATTERNS

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit

Enter your choice: 2

Enter the number of rows: 4

```
      1  
     1 2  
    1 2 3  
   1 2 3 4
```

RIGHT-ALIGNED PATTERNS

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit

Enter your choice: 3

Enter the number of rows: 4

```
* * * *  
 * * *  
  * *  
   *
```

RIGHT-ALIGNED PATTERNS

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit

Enter your choice: 1

Enter the number of rows: 1

```
*
```

RIGHT-ALIGNED PATTERNS

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle